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A candidate for lead-free ultrahigh-temperature piezoelectrics: the excellent electro-mechanical properties of Aurivillius oxides,



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Modified ultrahigh-temperature piezoelectric ceramics, $\text{Ca}_{1-5x}\text{Li}_{2x}\text{Nd}_{2x}\square_x\text{Bi}_2\text{Nb}_{2-2x}\text{Sc}_x\text{W}_x\text{O}_{9-1.5x}$ (CBNO-LiNd-ScW- x , $x = 0, 0.01, 0.02, 0.03, 0.04$; $\square$ represented A-site vacancies), were synthesized by a traditional solid-state reaction process. XRD Rietveld refinement, TEM, HRTEM and Raman spectroscopy were used to characterize the crystal structures and domain structures of the prepared ceramics. The lithium/neodymium (Li/Nd) and scandium/tungsten (Sc/W) co-substitution as well as the existence of small amounts of A-site vacancies significantly enhanced the ferroelectric and electro-mechanical properties of the CBNO-LiNd-ScW- x ceramics. The Curie temperature (T_c) decreased from 941 °C to 832 °C as the x values increased in CBNO-LiNd-ScW- x , because of an enhancement in the cation disordering between the A site and the bismuth oxide layer. The CBNO-LiNd-ScW-0.01 ceramic with an ultrahigh T_c of 908 °C had a high piezoelectric activity (d_{33}) of 20.6 pC N⁻¹, a large field-induced strain (S_{33}) of 0.27×10^{-3} and a high remnant polarization (P_r) of 6.1 $\mu\text{C cm}^{-2}$, due to the perfect growth in the ferroelectric domains and the contribution from the reversal of the defect dipoles ($V_{\text{Ca}}'' - V_{\text{O}}^{\bullet\bullet}$ and $\text{Sc}_{\text{Nb}}'' - V_{\text{O}}^{\bullet\bullet}$); interestingly, it still has a high d_{33} of 19.5 pC N⁻¹ after up to 800 °C thermal annealing. In addition, the CBNO-LiNd-ScW-0.01 ceramic had a relatively high resistivity of $>10^6 \Omega \text{ cm}$ at 600 °C and $>10^5 \Omega \text{ cm}$ at 700 °C.

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Introduction

Electromechanical transducing materials are required to sense strains, vibrations and noise under severe thermal conditions. Among the different types of acoustic and strain sensors, piezoelectrics are the best candidates when one considers sensitivity, cost and design.¹ Piezoelectric single crystals such as LiNbO₃, GaPO₄, La₃Ga₅SiO₁₄, Sr₂Nb₂O₇, YCa₄O(BO₃)₃, GdCa₄O(BO₃)₃ and A₂Ti₂O₇ (A = Nd and La) are generally used as high-temperature sensing components.^{2–5} However, the crystal growth of these materials needs expensive equipment and has a low yield. An important advantage of piezoelectric ceramics over single crystals is the ease with which the composition and properties may be tailored.² Aurivillius ferroelectrics (bismuth layer-structured ferroelectrics, BLSFs) have received much attention for their potential use in high-temperature piezoelectric devices, owing to their relatively high Curie temperature T_c .^{6–9} Modified and high-energy mechanochemical technique prepared bismuth titanate (Bi₄Ti₃O₁₂, BiT) ceramics are interesting for sensor

applications up to 500 °C.^{2,8} When an operating temperature is needed to be up to 800 °C, the choice of piezoelectric materials is limited. Among pure Aurivillius oxides, only two-layer CaBi₂Me₂O₉ and Bi₃TiMeO₉ (Me = Nb, Ta), have ultra-high T_c (>850 °C), between 870 and 943 °C. However, their piezoelectric coefficient (d_{33}) is relatively low, with values ≤ 10 pC N⁻¹,^{6,7,10–13} owing to the two-dimensional orientation restriction of the reversal of spontaneous polarization as well as the high coercive field (E_c). With a grain orientation technique of spark plasma sintering (SPS), the textured CaBi₂Nb₂O₉ (CBNO) ceramic had a higher d_{33} of 19.5 pC N⁻¹, which is nearly three times that of the ordinarily fired samples (OF).⁶ For two-layer Aurivillius oxides, the conductivity in the maximum d_{33} direction (ab plane) is much higher than that in the perpendicular axis (*i.e.*, along the c axis direction), which make the poling difficult.¹⁴ In addition, it is desirable to obtain a high d_{33} in BLSF ceramics produced by the ordinary firing method, from the viewpoint of commercial applications.

For BLSFs, A-site substitution can improve their piezoelectric properties and ferroelectric properties effectively, especially the co-substitution of alkali metal cations (Li⁺, K⁺, Na⁺, *etc.*) and lanthanon cations (La³⁺, Ce³⁺, Nd³⁺, Pr³⁺, *etc.*).^{15–18} With (Li, Ce) and (K, Ce) A-site co-substitution, the Ca_{0.85}(Li,Ce)_{0.075}Bi₄Ti₄O₁₅ and Ca_{0.85}(K,Ce)_{0.075}Bi₄Ti₄O₁₅ ceramics had d_{33} values of 20 and 17 pC N⁻¹, respectively,

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which are much higher than that of the pure $\text{CaBi}_4\text{Ti}_4\text{O}_{15}$ (5 pC N^{-1}).¹⁶ The modified $\text{Ca}_{0.9}(\text{NaCe})_{0.1}\text{Bi}_2\text{Nb}_2\text{O}_9$ ceramics obtained relatively high d_{33} , 13 pC N^{-1} , as compared to 5.8 pC N^{-1} of the pure CBNO. In our previous work, the LiNd co-substitution at the A site also induced an increase in the piezoelectric activity.¹⁹ On the other hand, an enhancement in the piezoelectric and ferroelectric properties can be caused by B-site substitution, *e.g.*, W substituted $\text{Na}_{0.5}\text{La}_{0.5}\text{Bi}_2\text{Nb}_2\text{O}_9$ and $\text{Bi}_4\text{Ti}_3\text{O}_{12}$, V substituted $\text{Bi}_{3.25}\text{Pr}_{0.75}\text{Ti}_3\text{O}_{12}$, and Sc^{3+} modified $\text{SrBi}_2\text{Ta}_2\text{O}_9$ ($\text{Me} = \text{Nb, Ta}$), *etc.*^{20–25} Therefore, it is expected that more excellent piezoelectric properties could be resulted from the A-site and B-site co-modification. The CeO_2 and Sc/Ta co-modified BTNO-based ceramics exhibited higher d_{33} of 18 pC N^{-1} , than that of the only Sc/Ta B-site modified BTNO ceramics (12 pC N^{-1}). In addition, the introduction of the A-site vacancies was recommended to enhance the ease of polarization.^{26,27}

Therefore, we selected A-site Li/Nd and B-site Sc/W co-modification as well as deliberately introducing small amounts of A-site vacancies, to improve the piezoelectric and ferroelectric properties of the CBNO materials, with the highest-known T_c (943°C) among the BLSFs. In this paper, modified $\text{Ca}_{1-5x}\text{Li}_{2x}\text{Nd}_{2x}\square_x\text{Bi}_2\text{Nb}_{2-2x}\text{Sc}_x\text{W}_x\text{O}_{9-1.5x}$ (CBNO-LiNd-ScW- x , $x = 0, 0.01, 0.02, 0.03, 0.04$; $\square$ represented A-site vacancies) ceramics were prepared by using a solid-state reaction process, and their structures, domain structures, excellent electro-mechanical properties and bulk conductivities were studied in detail.

Experimental

Appropriate amounts of starting the materials [Bi_2O_3 (99.5%), Nb_2O_5 (99.5%), CaCO_3 (99.8%), Nd_2O_3 (99.99%), Li_2CO_3 (99.99%), WO_3 (99.99%) and Sc_2O_3 (99.99%); Sinopharm Chemical Reagent Co., Ltd, CN] were mixed by a ball milling method in ethanol for 24 h. The mixtures were dried and then calcined at 950°C for 4 h. The calcined powders were remilled in ethanol for 24 h, dried, ground and cold isostatically pressed into pellets at 300 MPa. The pressed pellets were then sintered at 1150°C for 2 h, embedded in precursor powders. Electrodes for dielectric, impedance spectroscopies, and piezoelectric measurements were made of a fired-on platinum (Pt) paste, while silver (Ag) electrodes were used for the ferroelectric and electrostriction measures of the samples. Single phase CBNO-LiNd-ScW- x ceramics were determined from their powder X-ray diffraction (XRD) patterns (Fig. S1, ESI[†]), and all of the the diffraction peaks can be indexed to the orthorhombic $\text{CaBi}_2\text{Nb}_2\text{O}_9$ ($A2_1am$).⁶ Viewed from their scanning electron microscope (SEM) images, plate-like morphologies were observed for all samples, which are typical characteristics of Aurivillius ceramics due to the anisotropic nature of the crystal structure (Fig. S2, ESI[†]).

The Rietveld refinement for the powder diffraction data of the representative CBNO-LiNd-ScW-0.01 ceramic was carried out by the GSAS-EXPGUI program with orthorhombic space group, $A2_1am$ (Fig. S3, ESI[†]). The XRD data were collected in the 2θ range of $10\text{--}130^\circ$ with a step size of 0.02° and a step

time of 10 s at room temperature, by using an automated diffractometer (X'Pert PRO MPD, Philips, Eindhoven, The Netherlands) with a nickel filter ($\text{Cu K}\alpha$ radiation). The Raman spectra of all of the ceramics were obtained with an instrument (LabRAM HR800, Horiba Jobin Yvon, Lyon, France) in a backward scattering geometry (the exciting source was the 514.5 nm line from an argon ion laser). The lattice images of the ceramic samples were observed by using a transmission electron microscopy (TEM; Tecnai F30, FEI, Hillsboro, OR, USA) operated at 300 kV accelerating voltage. The temperature dependences of the dielectric properties of all of the samples were measured by using an LCR meter (4284A, Agilent, CA, USA). The ferroelectric properties and electrostriction properties of the samples were measured by using a ferroelectric analyzer (TF-2000, Aix ACCT, Aachen, Germany). After samples were poled in silicone oil under a dc electric field of $160\text{--}210 \text{ kV cm}^{-1}$ for 20–60 min at 160°C , their piezoelectric coefficient (d_{33}) was determined by using a piezo- d_{33} meter (ZJ-3B, Institute of Acoustics, Chinese Academic of Science, Beijing, China) at room temperature. Impedance spectroscopy (IS) of the samples was performed using an Impedance Analyzer (Solartron, SI 1260, Hampshire, U. K.) in the 0.1 Hz to 10 MHz frequency range at various temperatures.

Results and discussion

Phase transition and structural orthorhombicity

According to Table S1 (ESI[†]) derived from their XRD data, Fig. 1 demonstrates the evolution in lattice parameters (a , b and c) of the orthorhombic ($A2_1am$) CBNO-LiNd-ScW- x ($x = 0\text{--}0.04$) ceramics. The lattice parameters a and c gradually decrease while b increases continuously as the x value increases. In consequence, the orthorhombicity of the ceramics defined as $2(a - b)/(a + b)$,²⁸ reduces with increasing x values, from 7.6×10^{-3} to 5.4×10^{-3} ; thus, a relaxation in the orthorhombic distortion is induced (inset of Fig. 2a). In Fig. 2a, the $\text{Li}^+/\text{Nd}^{3+}$ and $\text{Sc}^{3+}/\text{W}^{6+}$ co-modification also has a strong influence on the T_c of CBNO-LiNd-ScW- x , and their T_c values gradually decrease from 941 to 832°C as the x value increases from zero to 0.04. In Fig. 2b, a low loss peak is just below T_c in pure CBNO and no obvious peak is observed for other

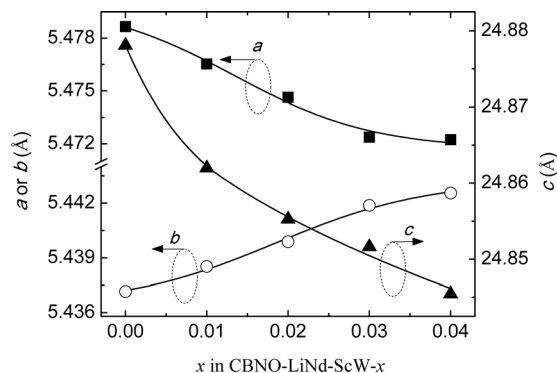


Fig. 1 The evolution in the lattice parameters of the CBNO-LiNd-ScW- x samples.

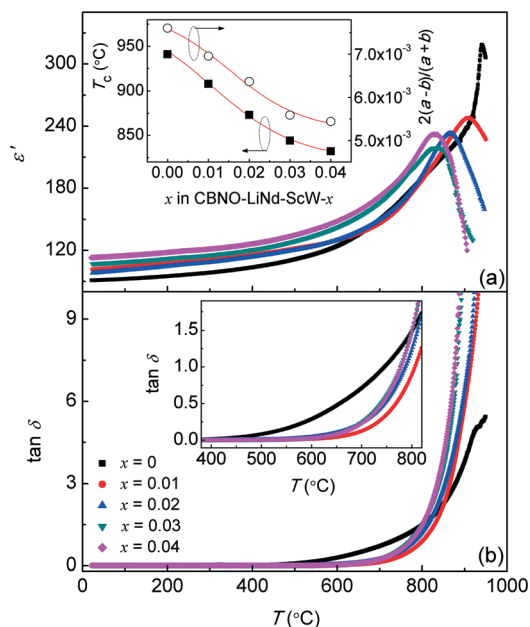


Fig. 2 The dielectric permittivity and dielectric loss of the CBNO-LiNd-ScW-*x* samples at 1 MHz. (a) Dielectric permittivity, with an inset showing the change trends in T_c and structural orthorhombicity $[2(a-b)/(a+b)]$ of the samples with an increase in the x value. (b) Dielectric loss, with an inset showing a regionally enlarged drawing.

samples. The modified CBNO-LiNd-ScW-*x* samples, especially CBNO-LiNd-ScW-0.01, have a lower loss ($\tan \delta$) in the higher temperature range (400–800 °C) (inset of Fig. 2b), which would be important for their high-temperature piezoelectric applications.

The ferroelectric to paraelectric phase transition temperature of the prepared samples should be closely related to their orthorhombic distortion. The structural distortion is explained in terms of a lattice mismatch between the BO_2 and AO planes in the perovskite-type unit of AB_2O_7 , and the tolerance factor (t) of the perovskite-layer units is given by:²⁹

$$t = (r_A + r_O) / \sqrt{2}(r_B + r_O) \quad (1)$$

where r_A , r_B , and r_O are ionic radii of an A-site cation, a B-site cation, and an oxygen ion, respectively. A decrease in t is resulted from the A-site substitution with smaller size $\text{Li}^+/\text{Nd}^{3+}$ (1.15 Å, 1.27 Å and 1.34 Å for Li^+ , Nd^{3+} and Ca^{2+} , respectively; 12 CN)^{16,30} and the B-site substitution with a larger average size (0.6725 Å and 0.64 for $\text{Sc}^{3+}/\text{W}^{6+}$ and Nb^{5+} , respectively; 6 CN).³⁰ The existence of the A-site vacancies could cause a further decrease in t . Commonly, a lower t suggests a higher degree of structural distortion. When only considering the perovskite-type unit, an enhancement in the lattice distortion is caused by the above co-modification (I). However, more A-site cations ($\text{Ca}^{2+}/\text{Nd}^{3+}$) by replacing the Bi^{3+} ions probably entered into the $(\text{Bi}_2\text{O}_2)^{2+}$ layers. Fig. 3 shows the Raman spectra of the CBNO-LiNd-ScW-*x* ceramics, and all of the samples exhibit the same phonon modes at about 69, 84, 103, 187, 210, 284, 304, 330, 454, 593 and 825 cm^{-1} , which are related to the

orthorhombic phase and are similar to those of the CBNO films reported by Simões *et al.*³¹ The modes at 84, 103 and 210 cm^{-1} are associated with the vibration of the A-site cations in the perovskite layers.^{32–34} A high-frequency shift for the mode at 84 cm^{-1} and a gradually vanishing mode at 103 cm^{-1} suggest a change in the vibration modes of the A-site cations. The mode at 825 cm^{-1} (E_g) originating from the vibration of the oxygen ions (O4, O5) in the Nb–O plane exhibits a slight low-frequency shift with an increasing x value. Also, the mode at 183 cm^{-1} in response to the vibration of the B-site ion (Nb/Sc/W) along the z direction, A_{1g} [Nb(z)], has a high-frequency shift.³⁵ Therefore, it is concluded that Li/Nd and Sc/W could be incorporated into the perovskite lattices and occupy the A and B site, respectively. Interestingly, a slight high-frequency shift is observed for the mode at 69 cm^{-1} [A_{1g} originating from Bi(z)], which originates from the Bi^{3+} ion vibration in the $(\text{Bi}_2\text{O}_2)^{2+}$ layers. This suggests that more A-site cations (Ca/Nd) are involved in the $(\text{Bi}_2\text{O}_2)^{2+}$ layers and the cation disordering between the A site and the $(\text{Bi}_2\text{O}_2)^{2+}$ layer is enhanced. This point is supported by the refined structure of CBNO-LiNd-ScW-0.01, where small amounts of Nd (0.42%) are introduced into the $(\text{Bi}_2\text{O}_2)^{2+}$ layers and more Ca (6.23%) occupies the fluorite layers (Table 1) as compared to 5.3% of the pure CBNO.³⁶ In addition, a broadened dielectric permittivity peak confirms the enhancement of the cation disordering (Fig. 2a). For BLSFs, the distortion of the perovskite layers is reinforced by the fifth strong bond Bi(2)–O(2) contracted by the Bi(2) of the $(\text{Bi}_2\text{O}_2)^{2+}$ layers with one of the apex O(2) of the perovskite layer, in addition to the four Bi(2)–O(3) bonds of the square pyramid configuration.¹⁰ The replacement of the polar Bi(2) (Bi^{3+}) with 6s² lone pair electrons by non-polar Ca(2)/Nd(2) ($\text{Ca}^{2+}/\text{Nd}^{3+}$) should weaken the Bi(2)–O(2) bond; consequently, the degree of the distortions of the octahedrons is decreased (II). It is likely that II has a stronger effect on the structural distortion than I. As a result, a reduction in the orthorhombicity is induced by the $\text{Li}^+/\text{Nd}^{3+}$ and $\text{Sc}^{3+}/\text{W}^{6+}$

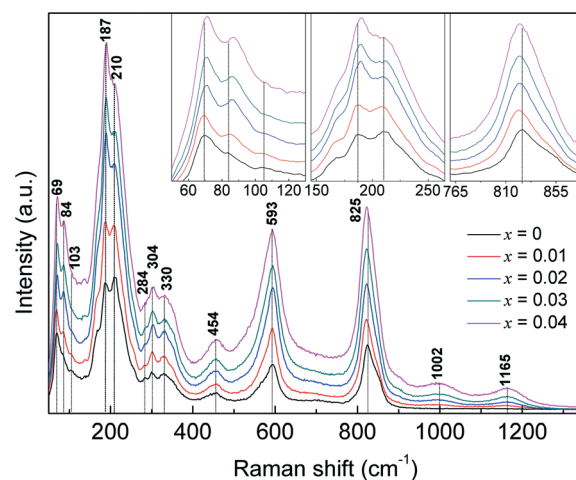


Fig. 3 The Raman spectra of the CBNO-LiNd-ScW-*x* samples at room temperature, with the insets showing the regionally enlarged drawings.

Table 1 The atomic coordinates, atomic occupancies and isotropic thermal parameters of the CBNO-LiNd-ScW-0.01 at room temperature^a

Atom	Site	x	y	z	100U _{iso}	Occupancy
Ca1	4a	0.2498(16)	0.7542(12)	0.5	0.04(7)	0.8245(17)
Li1	4a	0.2498(16)	0.7542(12)	0.5	0.04(7)	0.02
Nd1	4a	0.2498(16)	0.7542(12)	0.5	0.04(7)	0.012(4)
Bi1	4a	0.2498(16)	0.7542(12)	0.5	0.04(7)	0.132(5)
Bi2	8b	0.25	0.73052(23)	0.699791(30)	1.288(12)	0.9340(23)
Ca2	8b	0.25	0.73052(23)	0.699791(30)	1.288(12)	0.0628(9)
Nd2	8b	0.25	0.73052(23)	0.699791(30)	1.288(12)	0.0042(22)
Nb1	8b	0.2829(5)	0.2457(6)	0.58379(5)	0.421(32)	0.99
Sc1	8b	0.2829(5)	0.2457(6)	0.58379(5)	0.421(32)	0.005
W1	8b	0.2829(5)	0.2457(6)	0.58379(5)	0.421(32)	0.005
O1	4a	0.316(5)	0.171(4)	0.5	1.97(22)	0.99833
O2	8b	0.303(4)	0.3189(28)	0.6561(4)	1.97(22)	0.99833
O3	8b	0.476(5)	0.495(12)	0.2480(8)	1.97(22)	0.99833
O4	8b	0.540(5)	0.531(4)	0.5664(4)	1.97(22)	0.99833
O5	8b	0.6106(32)	0.024(4)	0.5839(4)	1.97(22)	0.99833

^a Lattice parameters: $a = 5.47656(10)$ Å, $b = 5.43861(10)$ Å, $c = 24.86232(28)$ Å. $Z = 4$. Reliability factors: $R_{wp} = 8.25\%$, $R_p = 6.27\%$, reduced $\chi^2 = 1.552$, and $D_{wd} = 1.371$. Space group: $A2_1am$.

co-modification, and a continuing drop in T_c is observed as the x value increases in CBNO-LiNd-ScW- x .

Ferroelectric polarization and structure

Ferroelectric hysteresis (P - E) loops of all of the samples are shown in Fig. 4a. A remnant polarization (P_r) of $2.7 \mu\text{C cm}^{-2}$ is observed for the pure CBNO, which is close to about $2.3 \mu\text{C cm}^{-2}$ reported by Yan, *et al.*⁶ With introducing $x = 0.01$ modification, the CBNO-LiNd-ScW-0.01 obtains a P_r of $6.1 \mu\text{C cm}^{-2}$, which is higher than twice that of the pure CBNO. However, higher concentration oxygen vacancies caused by higher level acceptor modification, could pin the domain walls and make the reversal of the domains difficult. This could cause the degraded ferroelectric properties of the CBNO-LiNd-ScW- x ($x = 0.02$ – 0.04) samples. Also, the degradation could be resulted from a dimensional mismatch between the perovskite and bismuth oxide layers due to a high concentration substitution.²¹ With attention, for the pure CBNO, the hysteresis loop goes towards saturation, having a concave region. But for the other acceptor CBNO-LiNd-ScW- x samples, the concave region in the P - E loops is missing and the loop is rather cigar-shaped.³⁷ This should be attributed to an increase in the leakage current due to the motion of the potential oxygen vacancies following the ferroelectric polarization.^{37,38} In order to ascertain the “pinning” effect of the oxygen vacancies on the domain walls, a comparison in the ferroelectric behavior after 30 switched cycles between the CBNO-LiNd-ScW-0.01 and CBNO-LiNd-ScW-0.03 samples is shown in Fig. 4b. It can be seen that both samples have a little increase in P_r in the initial stage. This could be attributed to a relaxation in the domain walls owing to the repeated polarization reversal. With further increasing the switching cycles, the P_r of CBNO-LiNd-ScW-0.03 decreases by degrees, while no drop is observed for the CBNO-LiNd-ScW-0.01 sample. The compared P - E loops before and after fatigue for the two samples are shown in the inset of Fig. 4b. Commonly, the defects (*i.e.*, oxygen vacancies) tend to agglomerate on the domain

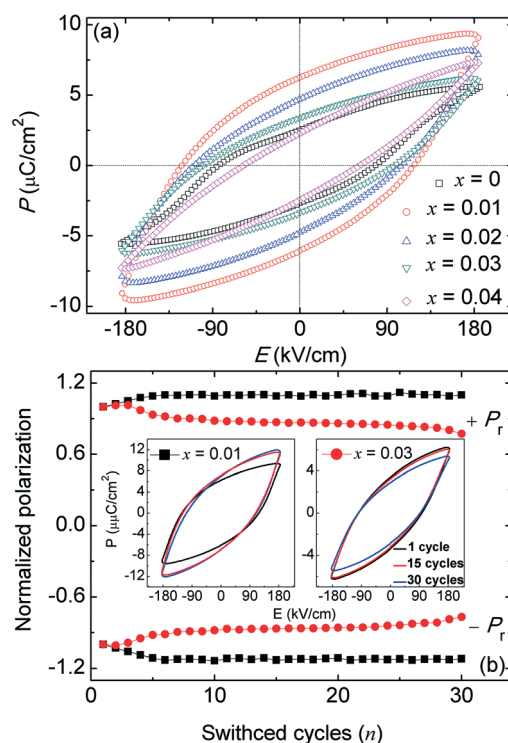


Fig. 4 (a) P - E loops of the CBNO-LiNd-ScW- x samples measured at 160 °C and 10 Hz. (b) A comparison in the ferroelectric behavior after 30 switched cycles between the CBNO-LiNd-ScW-0.01 and CBNO-LiNd-ScW-0.03 samples, with insets showing compared P - E loops after 1, 15 and 30 switched cycles.

walls during the switchable polarization process, and finally pin the domain walls.³⁹ An obvious degradation in the switchable polarization suggests that the defect “pinning” effect is pronounced in CBNO-LiNd-ScW-0.03.

The ferroelectric polarization (P_s) of two-layer Aurivillius oxides mainly originates from the displacements along the a axis from the corresponding positions in the parent tetragonal ($I4/mmm$). The polarization originated from the displacements along the b and c axes are cancelled because of the presence of

glide and mirror planes, respectively.⁴⁰ The total P_s of the displace-type ferroelectrics can be calculated by using Shimakawa's mode,⁴¹

$$P_s = \sum_i \frac{m_i \times \Delta x_i \times Qe}{V} \quad (2)$$

where m_i is the site multiplicity, Δx_i is the atomic displacement along the a axis from the corresponding position in the parent structure, Qe is the ionic charge for the i th constituent ion, and V is the volume of the unit cell. According to the refinement results (Table 1), the visualized crystal structure of CBNO-LiNd-ScW-0.01 and the contributions of each constituent ion to its total P_s are shown in Fig. 5a and c, respectively. Similar to $24.4 \mu\text{C cm}^{-2}$ of the pure CBNO,⁶ CBNO-LiNd-ScW-0.01 has an approximate P_s of $24.7 \mu\text{C cm}^{-2}$. Thus, the enhancement in the polarization in CBNO-LiNd-ScW-0.01 is not an intrinsic effect. According to Table S2 (ESI[†]), the A–O bond distances are in the range of 2.298(23) to 3.224(17) Å; therefore, a cation vacancy–oxygen vacancy dipole ($V_{\text{Ca}}'' - V_{\text{O}}''$) as well as an acceptor ion–oxygen vacancy dipole ($\text{Sc}_{\text{Nb}} - V_{\text{O}}''$) (Fig. 5b) are extremely probable in acceptor CBNO-LiNd-ScW- x

samples. These two types of defect dipoles could reorient with a high external field (180 kV cm^{-1}) by the aid of the movement of oxygen vacancies from one to another site in the oxygen octahedral framework. The cigar-shape-like P - E loop of acceptor CBNO-LiNd-ScW- x could result from the reversal of the defect dipoles following the ferroelectric polarization and their corresponding current density curves (J - E loops; Fig. S4, ESI[†]) confirm an increase in the switchable current due to the migration of oxygen vacancies. As a consequence, the reversal of the defect polarization contributes to the higher P_r and larger E_c (detected from the current peaks in J - E loops) in CBNO-LiNd-ScW-0.01. Furthermore, the excellent ferroelectric properties of the CBNO-LiNd-ScW-0.01 sample should be closely related to its perfect-growth macroscopic domains. A comparison in the domain morphologies between CBNO-LiNd-ScW-0.01 and pure CBNO is shown in Fig. 6. It can be observed from the dark field image using a superlattice reflection $\{[100]$ selected area electron diffraction (SAED), insets of Fig. 6a and c), that the ferroelectric domains of CBNO-LiNd-ScW-0.01 are present as strongly bright and dark contrasts (Fig. 6a and b), while there is a weak contrast in pure CBNO (Fig. 6c and d). This implies that the perfect growth in the ferroelectric domains is

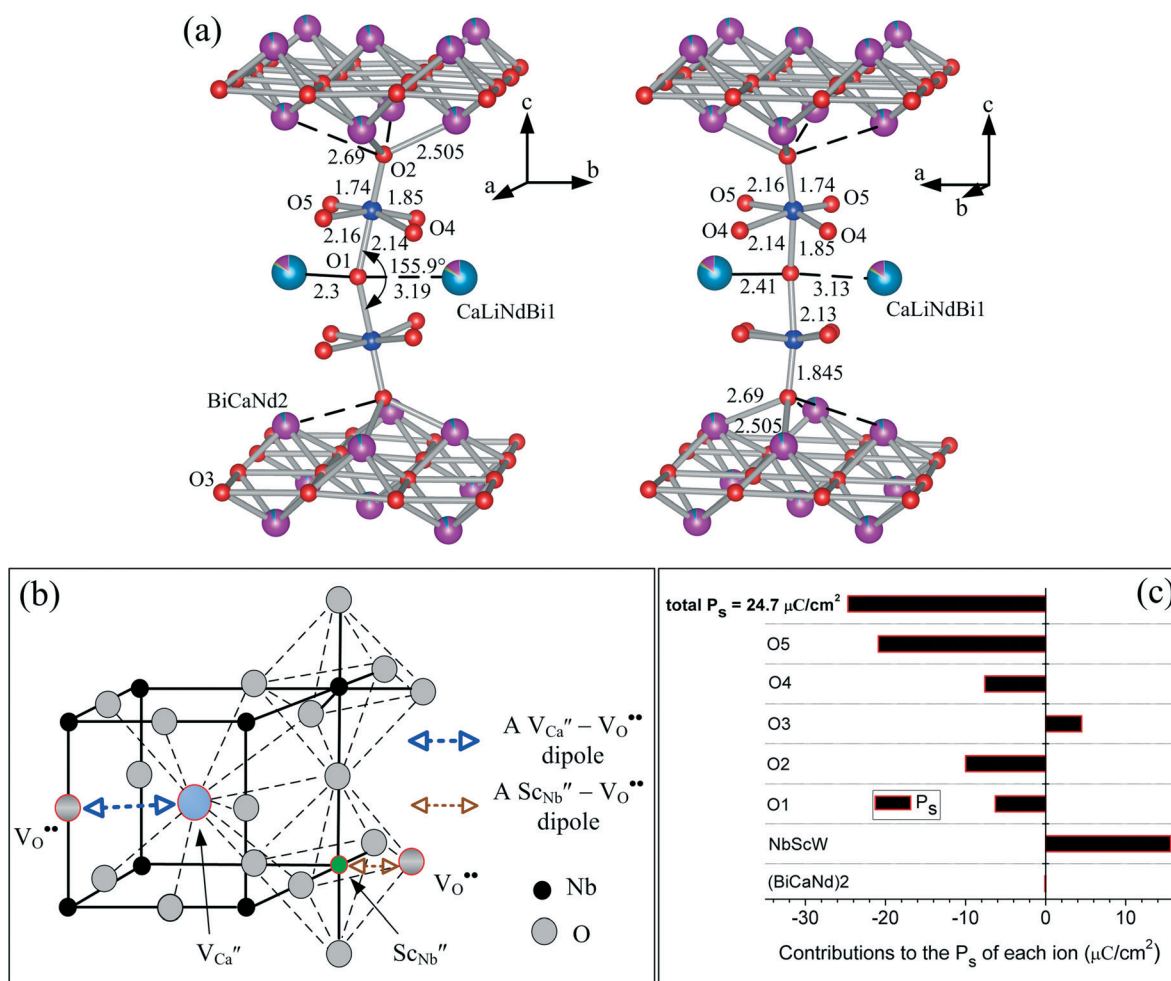


Fig. 5 (a) A visualization of the B_2O_7 ($\text{B} = \text{Nb/Sc/W}$) network distortion in the CBNO-LiNd-ScW-0.01 structure. (b) The defect dipoles' model for the acceptor CBNO-LiNd-ScW-0.01 structure. (c) The total spontaneous polarization (P_s) originating from each constituent ion in CBNO-LiNd-ScW-0.01 (the position of the A site on the a axis is fixed at the origin).

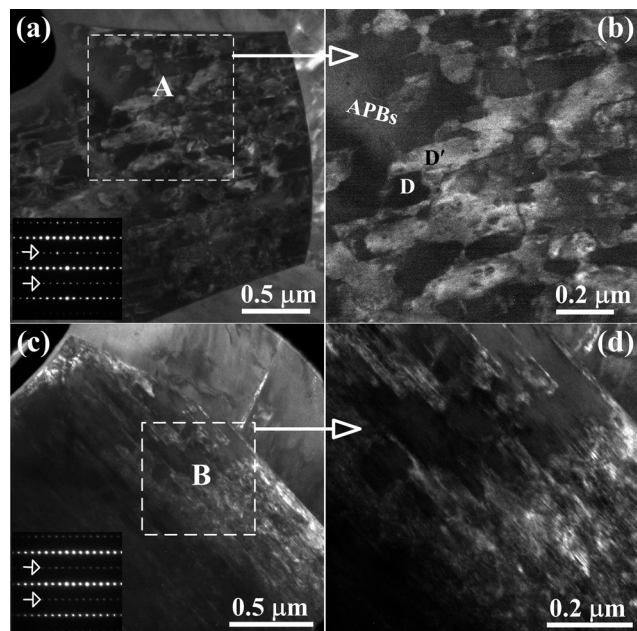


Fig. 6 (a) A dark field TEM image showing the domain structures for CBNO-LiNd-ScW-0.01, obtained from the superlattice reflections marked as arrows in the [100] SAED pattern (inset), (b) the enlarged experimental image of zone A in (a). (c) A dark field TEM image showing the domain structures for the pure CBNO, obtained from the superlattice reflections marked as arrows in the [100] SAED pattern (inset), (d) the enlarged experimental image of zone B in (c).

pronounced in CBNO-LiNd-ScW-0.01 while the CBNO has a lack of macroscopic domains. A discrepancy in the domain structures between them could determine their dissimilar ferroelectric and piezoelectric behaviors. In addition, detected from the domain region shown in Fig. 6b, two [001] zone-axis patterns which make an angle of 90° about the c axis are observed (inset of Fig. 7).⁴² This indicates that there are 90° domain walls existing in CBNO-LiNd-ScW-0.01. A corresponding [001] high-resolution transmission electron microscopy (HRTEM) image as well as a simulation as the inset, is shown in Fig. 7. The simulated image was performed with the *Jems* program using the *multislice* method,⁴³ with the refined crystallographic data. It is clear that the experimental and calculated contrasts are in fair agreement, which supports the refinement structure of CBNO-LiNd-ScW-0.01.

Electro-mechanical properties and stability

The bipolar and unipolar field-induced strains of the CBNO-LiNd-ScW- x samples are shown in Fig. 8. Commonly, Aurivillius materials have lower strain than other ferroelectrics, *e.g.*, PMN and $\text{Bi}_{0.5}\text{Na}_{0.5}\text{TiO}_3$, because of the two-dimensional orientation restriction of the reversal of their spontaneous polarization. Butterfly-typed strain curves are observed for all samples with an applied bipolar field, and these strain values are equivalent to those obtained from the unipolar field. With increasing x values, the field-induced strain values (S_{33}) first increase and then decrease in the CBNO-LiNd-ScW- x samples. In addition, the piezoelectric constants (d_{33}^*) of the samples

can be calculated directly from the slope of the strain *versus* the electric field curves,^{44,45} and their d_{33}^* values are in the range of $9.8\text{--}20.9\text{ pC N}^{-1}$. The change trend in d_{33}^* is consistent with that in S_{33} . Also, the piezoelectric activities (d_{33}) of the samples can be measured after dc field poling, and the obtained d_{33} values range from 9.7 ± 0.2 to $20.6 \pm 0.2\text{ pC N}^{-1}$ (Table 2). For each composition, it is interesting that its d_{33} value is very close to that of the d_{33}^* . Among them, the CBNO-LiNd-ScW-0.01 sample has the maximum strain value ($S_{33} = 0.27 \times 10^{-3}$) and piezoelectric activity ($d_{33}^* = 20.9\text{ pC N}^{-1}$ and $d_{33} = 20.6\text{ pC N}^{-1}$). The enhancement in the piezoelectric performance and electrostriction properties in CBNO-LiNd-ScW- x could be the defect polarization, similar to their ferroelectric properties. Also, the macroscopic domain walls' movement could contribute to piezoelectric activity. Table 2 lists a comparison of the piezoelectric properties among CBNO-LiNd-ScW- x and other reported ultrahigh-temperature Aurivillius ferroelectrics ($T_c > 850^\circ\text{C}$).^{6,9,18,25} It clearly demonstrates that the piezoelectric activity of CBNO-LiNd-ScW-0.01 can be comparable to those of the textured CBNO (19.5 pC N^{-1}) and modified $\text{Bi}_3\text{Ti}_{0.96}\text{Sc}_{0.02}\text{Ta}_{0.02}\text{NbO}_9 + 0.35\text{ wt\% CeO}_2$ (18 pC N^{-1}) ceramics, and much higher than those of other ultrahigh-temperature Aurivillius ferroelectric ceramics.

The effect of thermal depoling on the piezoelectric activities of all of the samples is shown in Fig. 9. The d_{33} value was measured at room temperature after annealing for 2h at each temperature (corresponding to the x -axis temperature). All of the samples demonstrate an excellent piezoelectric stability against temperature, except for a little drop at the low temperature range ($30\text{--}400^\circ\text{C}$). The trends are very similar to that of the pure CBNO reported by Yan *et al.*^{6,7} The monoclinic distortion of $\text{Bi}_4\text{Ti}_3\text{O}_{12}$ produces more types of

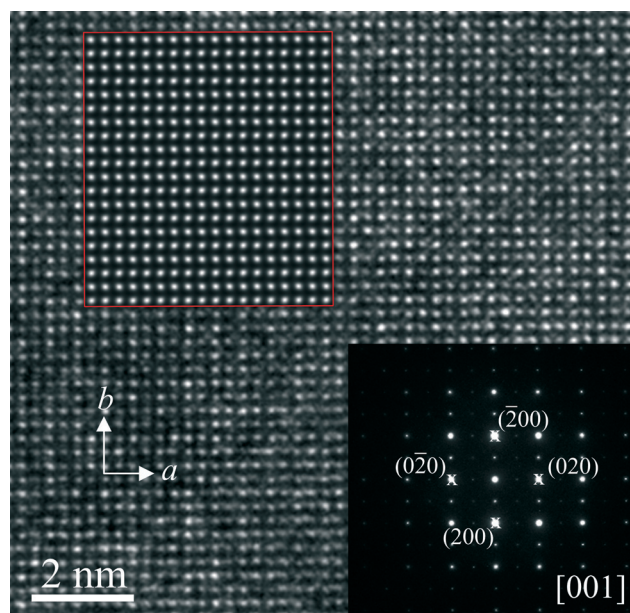


Fig. 7 The [001] HRTEM image and the corresponding [001] SAED pattern of CBNO-LiNd-ScW-0.01. The calculated image appeared as an inset (focus value $f = -90\text{ nm}$; crystal thickness $t = 5\text{ nm}$).

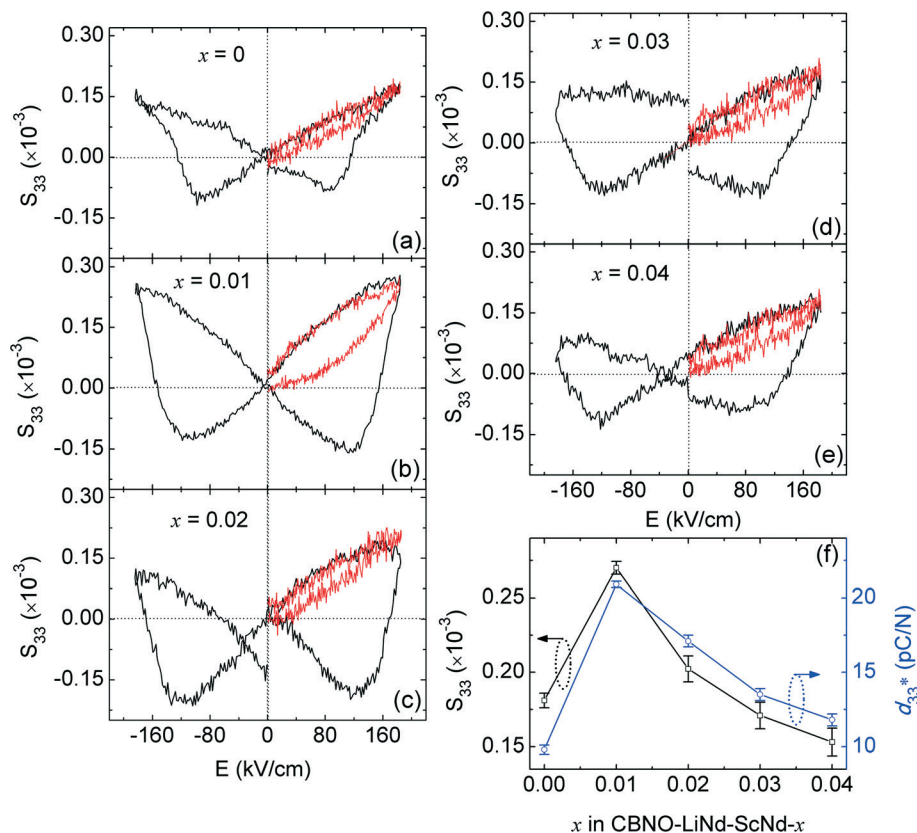


Fig. 8 (a–e) The electric-field-induced strain as a function of the bipolar (black line) and unipolar (red line) electric field of the CBNO-LiNd-ScW- x samples measured at 160 °C and 10 Hz. (f) The evolutions of field-induced strains (S_{33}) and piezoelectric activities (d_{33}^*) as the x value increases in CBNO-LiNd-ScW- x , d_{33}^* was determined from the slope of the strain versus the electric field curves in (a–e).

Table 2 A comparison of the properties of CBNO-LiNd-ScW- x ceramics and other reported ultrahigh-temperature Aurivillius ferroelectric ceramics ($T_c > 850$ °C)

Compound	T_c ($\pm 2^\circ$)	d_{33} (pC N $^{-1}$)	P_r (μ C cm $^{-2}$)	S_{33} ($\times 10^{-3}$)	d_{33}^* (pC N $^{-1}$)	$\tan \delta$ at 700 °C
CBNO-LiNd-ScW-0	941	9.7 ± 0.2	2.7	0.18 ± 0.005	9.8 ± 0.3	0.72
CBNO-LiNd-ScW-0.01	908	20.6 ± 0.2	6.1	0.27 ± 0.005	20.9 ± 0.2	0.12
CBNO-LiNd-ScW-0.02	873	17.3 ± 0.3	4.8	0.20 ± 0.01	17.1 ± 0.4	0.23
CBNO-LiNd-ScW-0.03	844	13.5 ± 0.3	3.4	0.17 ± 0.01	13.5 ± 0.4	0.25
CBNO-LiNd-ScW-0.04	832	11.4 ± 0.3	2.5	0.15 ± 0.01	11.8 ± 0.4	0.23
OF CBNO ⁶	943	7.5 ± 0.2	2.3 ± 0.3	—	—	—
SPS CBNO ⁶	940	19.5 ± 0.3	7.5 ± 0.5	—	—	—
Ca _{0.9} (NaCe) _{0.1} Bi ₂ Nb ₂ O ₉ ¹⁸	898	13	—	—	—	—
Ca _{0.95} La _{0.05} Bi ₂ Nb ₂ O ₉ ¹⁸	911	12.6	—	—	—	—
Bi ₃ TiNbO ₉ ⁹	913	6.2	—	—	—	—
Bi ₃ Ti _{0.99} W _{0.01} NbO ₉ ⁹	909	12	—	—	—	—
Bi ₃ Ti _{0.96} Sc _{0.02} Ta _{0.02} NbO ₉ ²⁵	905	12	—	—	—	—
Bi ₃ Ti _{0.96} Sc _{0.02} Ta _{0.02} NbO ₉ +0.35 wt %CeO ₂ ²⁵	900	18	—	—	—	—

the non-180° domain wall, and these non-180° domain walls are thermally unstable well below T_c , giving rise to the obvious d_{33} degradation against temperature.^{46,47} The orthorhombic CBNO has the maximum T_c among the Aurivillius oxides and the maximum orthorhombicity, $2(a-b)/(a+b)$, of 7.5×10^{-3} (7.6×10^{-3} in this work) which is higher than that of monoclinic Bi₄Ti₃O₁₂ (7.24×10^{-3}).^{28,36} A low-temperature depoling in CBNO-LiNd-ScW- x could be also attributed to the movements of the non-180° domain walls resulted from the potential monoclinic distortion. Meaningfully, CBNO-LiNd-ScW-0.01 still has a high d_{33} of 19.5 pC N $^{-1}$ after 800 °C

thermal depoling. For all of the samples, d_{33} decreases rapidly when the annealing temperature is near T_c , and becomes almost zero when the temperature is increased above T_c .

Electrical conduction mechanism

High-temperature resistivity is very important for Aurivillius piezoelectric high temperature applications. Fig. 10 shows the impedance spectra of the CBNO-LiNd-ScW-0.01 sample over 400–800 °C. Commonly, the electrode, grain boundary and grain components contribute to the conductivities of the

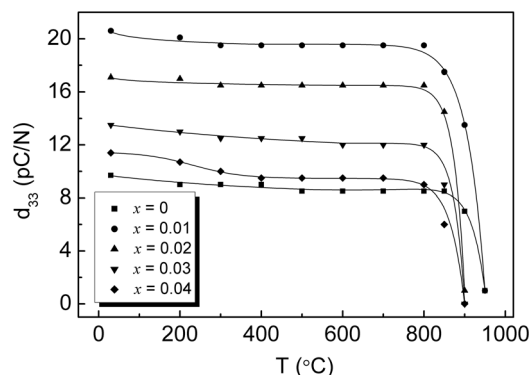


Fig. 9 The effect of the annealing temperature (30–950 °C) on the piezoelectric activity (d_{33}) of the poled CBNO-LiNd-ScW- x samples.

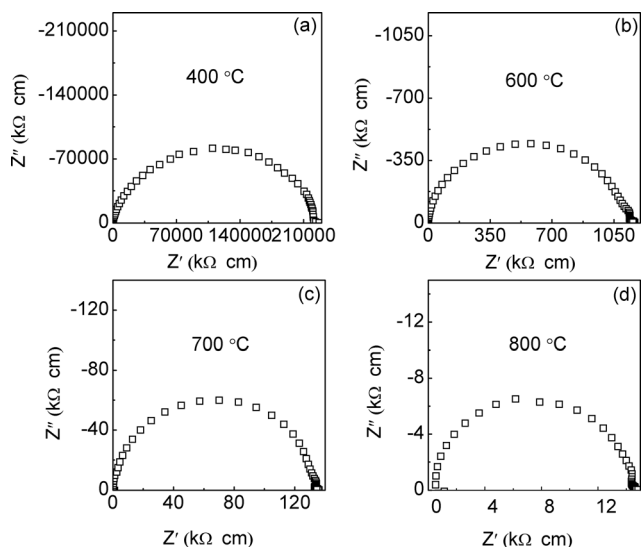


Fig. 10 The impedance spectra of the CBNO-LiNd-ScW-0.01 sample between 400 °C and 800 °C. (a) 400 °C, (b) 600 °C, (c) 700 °C and (d) 800 °C.

polycrystal ceramics and their effects on the conduction can be segregated into three parallel RC elements connected in series on an equivalent circuit, from low frequency to high frequency.⁴⁸ In Fig. 10c and d, only a single, essentially undistorted Debye-like semicircle is observed for CBNO-LiNd-ScW-0.01 at 700 and 800 °C. This indicates that the sample contains essentially one electrical component at the higher temperature range. Comparatively, a little squashed semicircle is obtained at a lower temperature (*e.g.*, 400 °C). According to Fig. S5 (ESI[†]), a lower capacitance plateau at a higher frequency range (10k–10M Hz) is observed in the corresponding capacitance (C') spectroscopic plots at each temperature, and there is an additional higher capacitance plateau (1–10k Hz) observed at lower temperatures. According to Li, *et al.*,⁴⁹ the lower capacitance plateaus of about 0.01 nF cm⁻¹ should be ascribed to the grain effect, whereas the high capacitance plateau is ascribed to the grain boundary effects. Correspondingly, only a single peak associated with the bulk conduction is observed in the high-temperature impedance (Z'') and complex modulus (M'') spectroscopic plots (Fig. S6, ESI[†]). However, two component peaks

(grain and grain boundary) can be seen in the low-temperature Z''/M'' versus $\log f$ plots (400 °C). Therefore, it is suggested that the grain and grain boundary both contribute to the low-temperature conductivity and capacitance, while the magnitudes of the conductivity and capacitance are mainly determined by the classic bulk response at a high temperature range.

Furthermore, the total resistivity (ρ) can be detected from the low-frequency intersects of the impedance complex plane data, and the variable-temperature resistivities of the CBNO-LiNd-ScW-0.01 sample are shown in Fig. 11. The resistivity of the sample decreases monotonously as the temperature increases. Meaningfully, the CBNO-LiNd-ScW-0.01 sample still has a relatively high resistivity of $>10^6 \Omega \text{ cm}$ at 600 °C and $>10^5 \Omega \text{ cm}$ at 700 °C, which would warrant applicability for high-temperature piezoelectric devices. In addition, the activation energy (E_a) for electrical conduction can be detected by using the Arrhenius equation,

$$\rho = \rho_0 \exp\left(\frac{-E_a}{kT}\right) \quad (3)$$

where E_a is the nominal activation energy per charge carrier and k is Boltzmann's constant. From eqn (3), it can be seen that $\log \rho$ has a linear relationship with $1/T$ and the E_a can be determined in terms of the slope of the Arrhenius plots. The CBNO-LiNd-ScW-0.01 sample has an E_a value of 1.73 eV at the higher temperature region (550–800 °C) and an E_a value of 1.31 eV at the lower temperature region (400–550 °C). It could be believed that the conduction mechanism at lower temperatures is an ionic and electronic mixed conduction while the intrinsic conduction is predominant at a higher temperature range.⁵⁰ The intrinsic E_a value (1.73 eV) could be close to the half energy of the band gap E_g ; thus, the CBNO-LiNd-ScW-0.01 sample has an E_g of approximately 3.46 eV, which is in good agreement with those of Bi₄Ti₃O₁₂, BaBi₄Ti₄O₁₅, Bi₃TiNbO₉ (BTNO)-based ceramics.^{50,51}

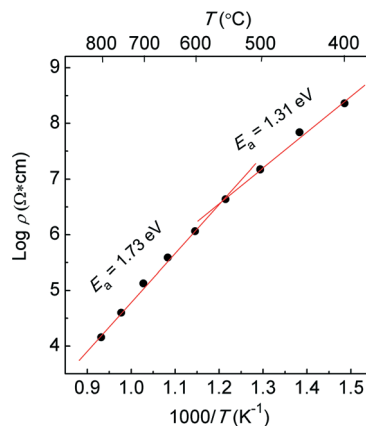


Fig. 11 The logarithmic total resistivities of the CBNO-LiNd-ScW-0.01 sample over 400–800 °C.

Conclusions

Modified ultrahigh-temperature piezoelectrics, $\text{Ca}_{1-5x}\text{Li}_{2x}\text{Nd}_{2x}\square_x\text{Bi}_2\text{Nb}_{2-2x}\text{Sc}_x\text{W}_x\text{O}_{9-1.5x}$ (CBNO-LiNd-ScW- x , $x = 0, 0.01, 0.02, 0.03, 0.04$; $\square$ represented A-site vacancies), were synthesized by a traditional solid-state reaction process, where $\text{Li}^+/\text{Nd}^{3+}$ and $\text{Sc}^{3+}/\text{W}^{6+}$ co-substituted the A-site Ca^{2+} and B-site Nb^{5+} , respectively, as well as the presence of small amounts of A-site vacancies. The structures and domain structures of the prepared ceramics were characterized by using XRD, SEM TEM, HRTEM and Raman spectroscopy. More A-site cations are involved in the $(\text{Bi}_2\text{O}_2)^{2+}$ layers, which led to a drop in the T_c in CBNO-LiNd-ScW- x , from 941 °C to 832 °C. A lower level modification ($x = 0.01$) gave rise to an enhancement in the electro-mechanical properties as well as the ferroelectric properties, due to the perfect growth in the ferroelectric domains and the contribution from the reversal of the defect dipoles ($\text{V}_{\text{Ca}}''-\text{V}_{\text{O}}^{\bullet\bullet}$ and $\text{Sc}_{\text{Nb}}''-\text{V}_{\text{O}}^{\bullet\bullet}$). With a higher level acceptor substitution, a degradation in d_{33} , S_{33} and P_r was observed in CBNO-LiNd-ScW- x ($x = 0.02-0.04$), which could be attributed to the “pinning” effect of the oxygen vacancies on the domain walls and a dimensional mismatch between the perovskite and bismuth oxide layer. The CBNO-LiNd-ScW-0.01 ceramic with an ultrahigh T_c of 908 °C had a high piezoelectric activity (d_{33}) of 20.6 pC N^{-1} , a large field-induced strain (S_{33}) of 0.27×10^{-3} and a high P_r of $6.1 \mu\text{C cm}^{-2}$; interestingly, it still has high d_{33} of 19.5 pC N^{-1} after up to 800 °C thermal annealing. In addition, the CBNO-LiNd-ScW-0.01 ceramic had relatively high resistivity of $>10^6 \Omega \text{ cm}$ at 600 °C and $>10^5 \Omega \text{ cm}$ at 700 °C. Therefore, the CBNO-LiNd-ScW-0.01 ceramic would be a promising candidate for use as an ultrahigh-temperature piezoelectric.

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